# Huawei Access Node QoS Congestion Event Root Cause Analysis and Resolution SOP

**Node:** BAS-ACC-53 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_4445

## Description

On Huawei Access nodes, QoS congestion events are primarily caused by the interaction between the egress queue management (EQM) algorithm and the microburst mitigation mechanism. When microbursts occur, the EQM algorithm attempts to prevent tail drops by dropping packets earlier in the queue, but this can lead to QoS queue starvation, especially in scenarios where the ingress rate significantly exceeds the egress capacity. Furthermore, the Huawei Access node's implementation of the 802.1Qaz standard for traffic policing and shaping can exacerbate the issue, causing the interface to experience congestion and subsequent tail drops. As a result, the QoS congestion event is triggered, and the ALARM_4445 is raised. In this SOP, we will perform a comprehensive analysis of the root causes and implement corrective measures to mitigate the issue.

## Pre-requisite Checks

- Verify that the Basavanagudi POP has been physically checked by the on-site engineer to ensure that all cabling and connections are secure.

- Check the redundant link status to ensure that the primary link is functioning correctly and the secondary link is ready to take over in case of a failure.

- Confirm that the Huawei Access node (BAS-ACC-53) is operational and online.

## Potential Root Causes

| Cause | Probability |
|---|---|
| Microbursts on the egress interface | High |
| QoS queue starvation due to EQM algorithm | Medium |
| Traffic policing and shaping configuration issues | Low |

## Diagnostic Commands & Expected Outputs

```
Command: display cpu-intf queue, Expected Output: Queue statistics for the egress interface, including packet drops and queue lengths., Description: This command provides detailed statistics for the egress interface, allowing us to identify potential issues with packet drops and queue lengths, which may indicate QoS queue starvation.

Command: display interface traffic-statistics, Expected Output: Traffic statistics for the egress interface, including ingress and egress rates, packet counts, and error rates., Description: This command provides traffic statistics for the egress interface, enabling us to analyze the ingress and egress rates, packet counts, and error rates, which may reveal microburst-related issues.
```

```
Command: display qos police stat, Expected Output: Traffic policing and shaping statistics, including packet drops and rate limits., Description: This command provides statistics for the traffic policing and shaping configuration, allowing us to identify potential issues with packet drops and rate limits, which may indicate configuration-related problems.
```

## Resolution Steps

Step 1: Step: Verify and adjust the traffic policing and shaping configuration, Command: undo qos police, Description: This command removes the traffic policing and shaping configuration. If the issue is configuration-related, this should resolve the problem.

Step 2: Step: Adjust the EQM algorithm parameters, Command: system-view interface GigabitEthernet0/0/0 queue-mode strict-priority queue-number 1 queue-threshold 100 queue-limit 200, Description: This command adjusts the EQM algorithm parameters to prevent QoS queue starvation. The specific parameters will depend on the node's configuration and the nature of the microbursts.

Step 3: Step: Monitor and adjust the QoS configuration, Command: display qos config, Description: This command displays the current QoS configuration, allowing us to verify that the adjustments made in the previous steps are effective.

## Post-Check Verification

```
Command: display cpu-intf queue, Expected Output: Queue statistics for the egress interface, indicating no packet drops or queue lengths., Description: This command provides detailed statistics for the egress interface, allowing us to verify that the issue has been resolved.

Command: display interface traffic-statistics, Expected Output: Traffic statistics for the egress interface, indicating no microburst-related issues., Description: This command provides traffic statistics for the egress interface, enabling us to analyze the ingress and egress rates, packet counts, and error rates, which should be within normal limits.

Command: display qos police stat, Expected Output: Traffic policing and shaping statistics, indicating no packet drops or rate limits., Description: This command provides statistics for the traffic policing and shaping configuration, allowing us to verify that the adjustments made in the previous steps are effective.
```

## Roll-back Procedures

- Step: Restore the original traffic policing and shaping configuration, Command: qos police, Description: This command restores the original traffic policing and shaping configuration, effectively rolling back the changes made in the previous steps.

- Step: Revert the EQM algorithm parameters, Command: undo system-view undo interface GigabitEthernet0/0/0 undo queue-mode strict-priority undo queue-number 1 undo queue-threshold 100 undo queue-limit 200, Description: This command reverts the EQM algorithm parameters to their original values, effectively rolling back the changes made in the previous steps.